

Unit 13, Lesson 3: Determining Appropriate Measures of Center and Variation

Benchmark

MA.7.DP.1.1 Determine an appropriate measure of center or measure of variation to summarize numerical data, represented numerically or graphically, while taking into consideration the context and any outliers.

Learning Target

- ☐ I can determine an appropriate measure of center and variation to summarize numerical data.

13.3.1 Warm-Up: Between Two Numbers

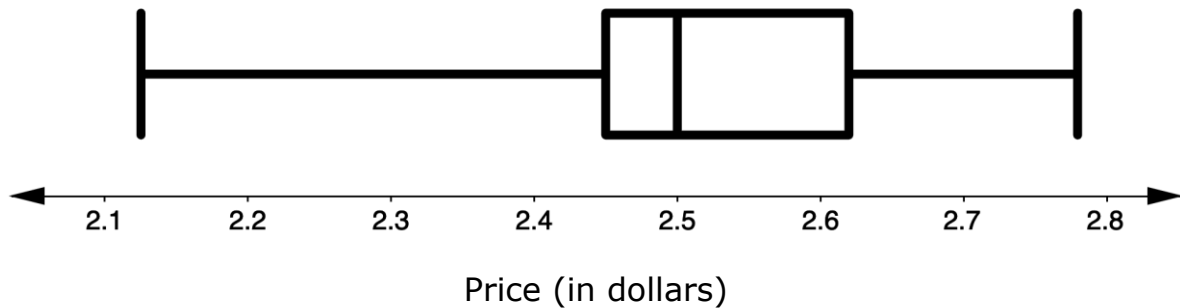
The temperature of the sun's core is 15 million degrees Celsius.

The temperature of Earth's core is 6000 degrees Celsius.

1. What is the ratio of core temperatures of the sun to Earth?

13.3.2 Guided Practice: Measures of Center and Variation

1. The U.S. Energy Information Administration reports weekly average gas prices for the United States. The box plot represents the weekly average gas prices in 2019.

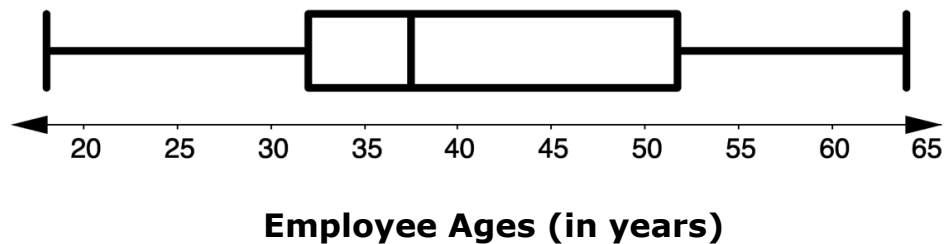
2019 Weekly Gas Prices

- Explain which measure of center, the mean or the median, best describes the center of the data.
- If possible, find the value of the measure you chose in Part A.
- Describe the variation of the weekly average price of gasoline.
- Given your description, does the interquartile range or range better represent the variation in price?

2. Robin, a small business owner, is trying to determine the cost of providing a small group health care plan to her 30 employees. Her insurance agent asks her for a summary of the employees' ages, since age is a factor that affects the premium cost of insurance. The employees' ages are displayed in the stem-and-leaf plot and box plot.

Stem	Leaf
1	8, 9
2	5, 8
3	0, 0, 1, 2, 2, 6, 6, 6, 7, 7, 7, 8, 9
4	0, 1, 2, 8
5	1, 2, 5, 8, 8, 9, 9
6	0, 4

Key: 1|8 = 18



Complete the statements to explain which measure of center and variation Robin should provide to her insurance agent to accurately summarize the ages of her employees.

- a. The ☐ mean
☐ median best summarizes the center of the data for the agent because ...
- b. The ☐ IQR
☐ range best summarizes the variation of the data for the agent because ...

13.3.3 Try It: Measures of Center and Variation

1. Tom Brady is considered by most to be one of the best NFL quarterbacks of all time. His pass-completion percentages from 2009 to 2019 are shown in the box plot and stem-and-leaf plot.

Tom Brady's Completion Percentages 2009-2019

center of

☐ mean

☐ median

the data. The

☐ mean

☐ median

is not

the best measure

☐ 60.5% and 60.8%

☐ 66.3% and 67.4%

of center because it is influenced by the values

that are far from the rest of

the data values.

Stem	Leaf
60	5, 8
61	
62	
63	0
64	1, 4
65	6, 7, 8, 9
66	3
67	4

Key: 60|5 = 60.5%

- b. The ☐ IQR
☐ range best summarizes the variation of the data. The ☐ IQR
☐ range is not the best measure of variation because it is affected by the value ☐ 60.5%
☐ 67.4% that is far from the rest of the data.

2. The Rachards family is moving to Gainesville, Florida and want to purchase a home in a specific area. They have a budget of \$160,000. The prices of homes in the area are shown below, in thousands of dollars.

150, 302, 201, 212, 150, 160, 140, 145

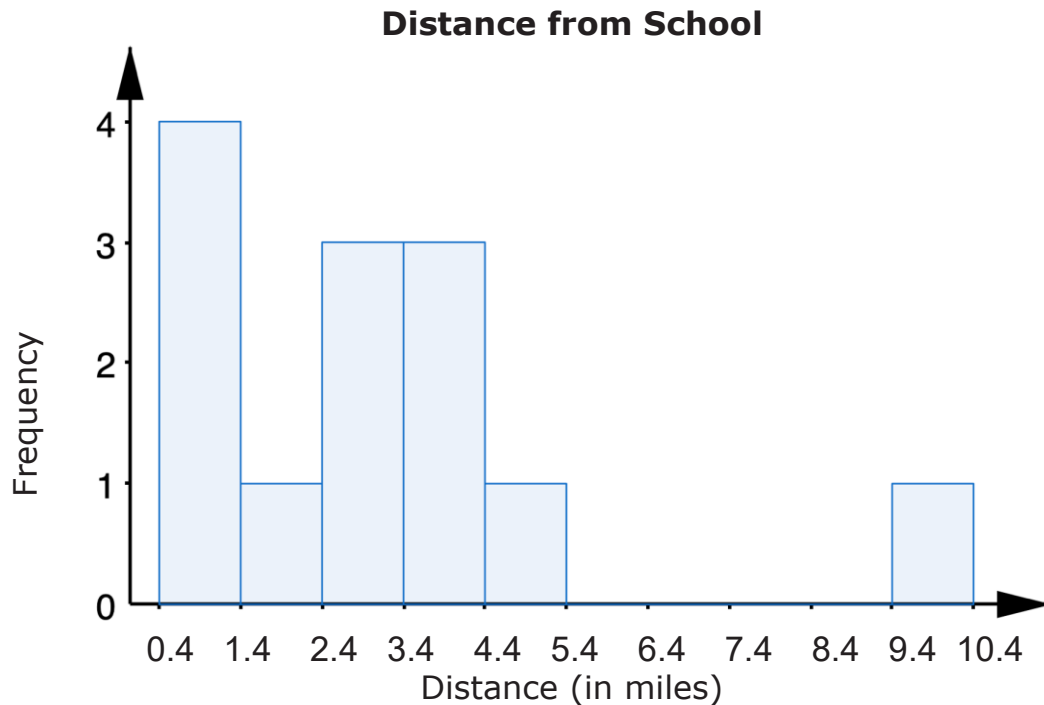
The Rachards are concerned that the homes may be out of their budget, so they ask the realtor to summarize the home prices.

- a. Which measure of center should the realtor report to the Rachards, the mean or the median? Explain your reasoning.
- b. Which measure of variation should the realtor report to the Rachards, the interquartile range or the range? Explain your reasoning.

13.3.4 Your Turn!

Zane and 12 of his friends listed how far, in miles, they live from the school and created a histogram.

0.4, 0.5, 1, 1, 2.4, 2.7, 3, 3.2, 4, 4.2, 4.4, 5, 10.3



1. Zane concluded that the best description of the typical distance that a student in his class lives from school is 3 miles, the value of the median.

Do you agree with Zane? Explain your reasoning.

2. Zane explained to his classmate, Shelby, that the students lived anywhere from 0.4 to 10.3 miles from school, so the range of 9.9 miles was the best summary for the spread of the data.

Shelby argued, "That's misleading, only one person lived more than 5.4 miles from school, so the interquartile range is a more accurate depiction of the spread of the distances."

- a. Do you agree with Zane or Shelby? Explain your reasoning.
 - b. Under what circumstances would you agree with the other person's chosen measure of variation?
3. Noelle is Zane's friend who used to live 10.3 miles away from the school. She recently moved closer, and now only lives 5.5 miles away from the school.

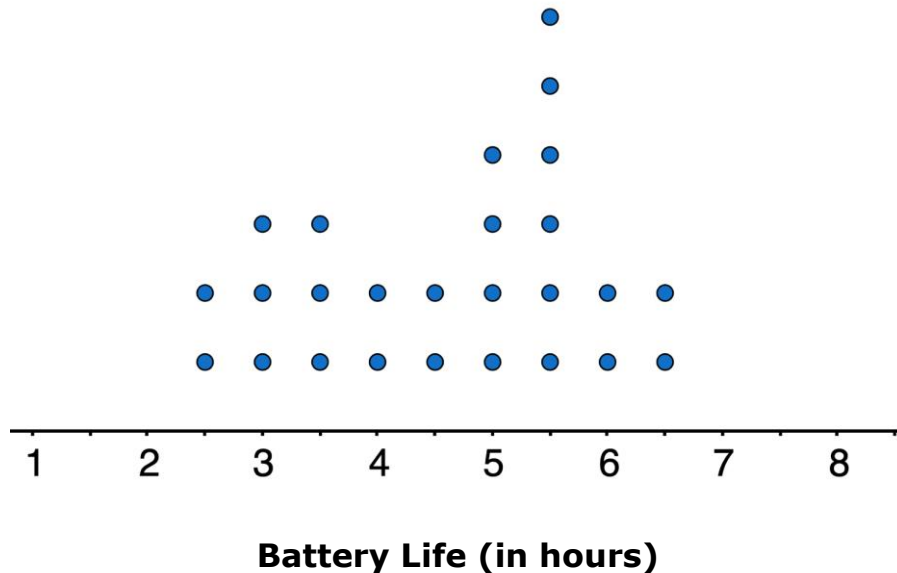
The data set with Noelle's new distance from school, in green, is shown.

0.4, 0.5, 1, 1, 2.4, 2.7, 3, 3.2, 4, 4.2, 4.4, 5, 5.5

- a. What measure of center best describes the typical distance Zane's friends live from school? Explain your reasoning.
- b. What measure of variation best describes the spread of the data distribution? Explain your reasoning.

13.3.5 Wrap-Up: Ticket Out the Door

The dot plot summarizes the hours of battery life for students' cell phones when streaming videos.



Mr. Funk's world history class has an upcoming field trip to Tallahassee, Florida that involves a five-hour drive. The students want to be able to watch videos on their phones during the drive without charging their cell phones.

1. Determine the appropriate measure of center and variation for the battery life of the students' cell phones when streaming videos.

